

Homework 1

Math 2031 — Professor Hamilton

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Section 1.1

Question 4

- (A) $\exists$ triangle T such that T is not an isosceles triangle.
- (B) $\forall$ even numbers N , there does not exist an $A \in \mathbb{R}$ such that $N = 3A$.
- (C) There exists a door in the building that was not locked.
- (D) There exists a new car such that there is nothing wrong with it.
- (E) All angles of a triangle are less than or equal to 90 degrees.
- (F) $\forall$ sets S , S does not contain infinitely many elements.

Question 5

- (A) $\forall x \in \mathbb{R}, x^2 + x + 1 \neq 0$.
- (B) $\exists x \in \mathbb{R}$ such that $x \geq 100$.
- (C) f is a polynomial function and f is not continuous at 0.
- (D) f is a polynomial function and f is not continuous anywhere.
- (E) $\exists x \in \mathbb{R}$ such that $\forall y \in \mathbb{R}, y \neq x^3$.
- (F) $\forall$ real-valued functions $f(x)$, $f(x)$ is continuous at any real number x .

Question 7

Conjecture: The sum of two even integers is divisible by 4.

- (A) For all integers x, y , if x and y are even then $x + y$ is divisible by 4.
- (B) x, y are even integers and $x + y$ is not divisible by 4.
- (C) Let $x = 2$ and $y = 0$. In this case $x + y = 2 + 0 = 2$, which is not divisible by 4. Therefore the conjecture is false; the statement would only be true for even integers x, y that are greater than or equal to 2.

Section 1.2

Question 5

- (A) x and y are real numbers such that $xy = 0$ and $x \neq 0$ and $y \neq 0$.
- (B) $\exists x \in \mathbb{Z}$, x^2 is not odd or $x^3 - 1$ is not divisible by 4.
- (C) $\exists n \in \mathbb{Z}$ such that $\forall k \in \mathbb{Z}$, $n \neq 2k \wedge n \neq 2k + 1$.
- (D) $\forall r \in \mathbb{R}$, $r \leq 1 \wedge r \geq 2$.
- (E) $\forall r \in \mathbb{R}$, r is less than or equal to 2 and less than or equal to 1.
- (F) All functions are differentiable at 0 and continuous at 0.

Question 6

Conjecture: Every multiple of 6 is even and is not a multiple of 4.

- (A) $\forall n = 6a \mid n = 2k \wedge n \neq 4b$. $\{n, a, k, b \in \mathbb{N}\}$.
- (B) $\exists n = 6a \mid n \neq 2k \vee n = 4b$. $\{n, a, k, b \in \mathbb{N}\}$
- (C) Let's assume that $n = 6a \mid n, a \in \mathbb{N}$, i.e, n is an integer multiple of 6. Our goal is to establish that every $n \in \mathbb{N}$ is divisible by 2, ergo even, and not every n is a multiple of 4, i.e, $n \neq 4b, b \in \mathbb{N}$. Let us rewrite n by factoring a 2, in which case we get $n = 2(3a)$. Here $3a$ is an integer and thus we have expressed n as integer multiple of 2, $\therefore$ by definition it is even. Next, we want to prove that not every n is a multiple of 4, which here we will prove by contradiction. Let n be a multiple of 6 and 4. Thus $6a = 4b$, remember a and b are both integers, however if we divide the RHS by 4 or the LHS by 6, we get following fractions; $a = \frac{4}{6}b$ or $b = \frac{6}{4}a$ which contradicts the operations of the integers, in that they are closed under multiplication. Since the negation leads to contradiction, the original statement must be true. **QED**

Section 1.3

Question 1

- (A) $P : \forall \text{ Hexagons } H, H \text{ has 6 sides.}$
 $\sim P : \exists H \mid H \text{ does not have 6 sides.}$
- (B) $P : \forall n, a, b \in \mathbb{N} \ n = 2a \vee n = 2b + 1 .$
 $\sim P : \exists n, a, b \in \mathbb{N} \mid n \neq 2a \wedge n \neq 2b + 1.$
- (C) $P : \forall f \text{ that is differentiable at 0, } f \text{ is continuous at 0.}$
 $\sim P : \exists f \text{ that is differentiable at 0 such that } f \text{ is not continuous at 0.}$
- (D) $P : \forall x \in \mathbb{R}, \exists y \in \mathbb{R} \text{ such that } \sqrt{x} = y.$
 $\sim P : \exists x \in \mathbb{R} \text{ such that } \forall y \in \mathbb{R}, \sqrt{x} \neq y$